Appendix B

Calculating Subnets

Chapter 8 provides context for how subnetting works and how subnets contribute to network management and security. If you need additional details on how to calculate subnets, this appendix guides you through these calculation processes.

Subnetting Questions on Exams

Although it's impossible to know for sure, you're likely to encounter two types of subnet calculation problems on some networking exams:

- Given certain network requirements (such as required number of hosts or required number of subnets), calculate possible subnets and host IP address ranges.
- Given an IP address, determine its subnet's network ID, broadcast address, and first/last host addresses.

Consider the following example of a network that initially contains only a single class C subnet, and all its hosts (up to 254) are on the same subnet. By borrowing only one bit from the host ID, you can have two subnets with up to 126 hosts (a more manageable number) per subnet. Let's step through how this works.

IPv4 Subnet Calculation in Binary

Suppose you have a network, as illustrated in Figure B-1, with one router and a switch connected to one interface on the router with your endpoint devices connected to that one switch. You then add a second switch to a different interface on the router to divide your local network into two LANs—each router interface connects to a single LAN. You can then redistribute your network devices so some of them stay connected to the LAN on the first switch, and others are moved to the new LAN on the second switch (Figure B-2).

Figure B-1 Network with one switch (one LAN) connected to one router

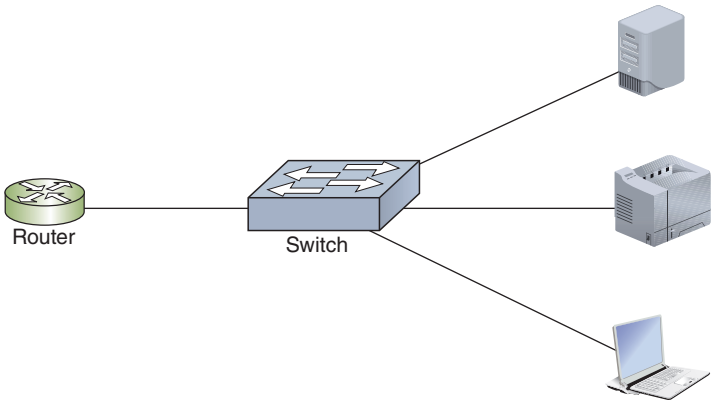
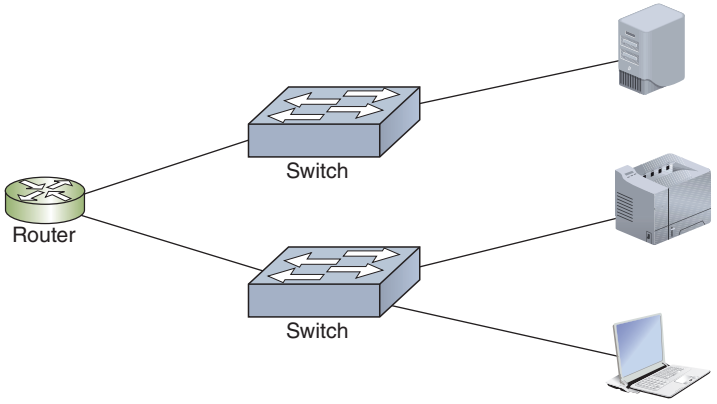


Figure B-2 Network with two switches (two LANs) connected to one router



Note B-1 | In this example and in many others in this appendix, a **bold red font** is used for the network ID portion of an IP address.

The network ID of the original network is 192.168.89.0, and its subnet mask is 255.255.255.0. From this IP address block, let's create two subnets of IP addresses, one for each LAN:

1. **Borrow from host bits**—Initially, the network ID is 24 bits. First convert the network ID to binary:
   - Network ID 192.168.89 in binary:
     11000000.10101000.01011001

     Borrow one bit from the host portion to give to the network ID, which will then have 25 bits (notice one additional red underlined bit in the fourth octet). Here, the borrowed bit is an "x" to show it can have a value of 0 or 1:
   - **11000000.10101000.01011001.x**

     How many subnets do you have now? The underlined red bit can be either a 0 or a 1, which gives you the possibility of two subnets:
   - **11000000.10101000.01011001.0**
   - **11000000.10101000.01011001.1**
2. **Determine the subnet mask**—Recall that the subnet mask marks the bits in an IP address that belong to the network ID. Therefore, the subnet mask for both subnets is as follows:
   - **11111111.11111111.11111111.1**0000000 or decimal 255.255.255.128

     To calculate that last octet, you convert binary 10000000 to decimal, which is 128. You can use a calculator to do the conversion, manually calculate it, or memorize this and a handful of other common binary-to-decimal conversions.
3. **Determine the network IDs**—Recall that in the network ID, the underlined red bit can be a 0 or 1—these two possibilities yield two subnets. Therefore, the network ID for each subnet is as follows:
   - Subnet A: **11000000.10101000.01011001.0**0000000 or decimal 192.168.89.0
   - Subnet B: **11000000.10101000.01011001.1**0000000 or decimal 192.168.89.128

     In CIDR notation, which identifies the first 25 bits as the network portion, the network ID for each subnet is the following:
   - Subnet A: 192.168.89.0/25
   - Subnet B: 192.168.89.128/25
4. **Determine the ranges of IP addresses for hosts in the subnet**—Start with the range of available IP addresses for subnet A. For host addresses, use the last seven bits in the last octet. (The first bit

for this octet in Subnet A is always 0 and belongs to the network ID.) Start counting in binary and convert to decimal:

- 00000000 is not used because it's the network ID for this subnet
- 00000001 or decimal 1
- 00000010 or decimal 2
- 00000011 or decimal 3
- ...
- 01111110 or decimal 126
- 01111111 or decimal 127, which is reserved for broadcasting within the subnet and cannot be used as a host address

The range of possible IP addresses is 192.168.89.0 through 192.168.89.127 (which is 128 possibilities). However, the first and last addresses cannot be used. Therefore, the range of host IP addresses for subnet A is 192.168.89.1 through 192.168.89.126 (that's 128–2, yielding 126 possibilities).

For subnet B, the first bit of the last octet is always 1, and the range of host addresses is as follows:

- 10000000 is not used because it's the network ID for this subnet
- 10000001 in decimal: 129
- 10000010 in decimal: 130
- 10000011 in decimal: 131
- ...
- 11111110 in decimal: 254
- 11111111 in decimal: 255, which is reserved for broadcasting within the subnet and cannot be used as a host address

Therefore, the range of host IP addresses for subnet B is 192.168.89.129 through 192.168.89.254 (another 126 possibilities).

The results of each of the preceding steps are summarized in Table B-1.

Table B-1 Steps to divide IP addresses for network ID 192.168.89.0 into two subnets

Step 1: Borrow from host bits.				
Network ID	192	168	89	0
In binary	11000000	10101000	01011001	00000000
Borrow 1 bit	11000000	10101000	01011001	x0000000
Step 2: Determine the subnet mask.				
In binary	11111111	11111111	11111111	10000000
In decimal	255	255	255	128
Step 3: Determine the network IDs.				
Network ID A	11000000	10101000	01011001	00000000
In decimal	192	168	89	0
In CIDR notation	192.168.89.0/25			
Network ID B	11000000	10101000	01011001	10000000
In decimal	192	168	89	128
In CIDR notation	192.168.89.128/25			

Continues

Step 4: Determine range of host IP addresses.				
Subnet A:				
First host, binary	11000000	10101000	01011001	000000001
First host, decimal	192	168	89	1
Last host, binary	11000000	10101000	01011001	01111110
Last host, decimal	192	168	89	126
Subnet B:				
First host, binary	11000000	10101000	01011001	10000001
First host, decimal	192	168	89	129
Last host, binary	11000000	10101000	01011001	11111110
Last host, decimal	192	168	89	254

Notice that you gained a network (the *one* larger network became *two* smaller subnets), but you lost some host addresses (each subnet offers a possible 126 host addresses for a total of 252 hosts, which is two fewer than the original network's possible 254 hosts). The difference is because each subnet needs its own network ID and broadcast address. Instead of the original network's one network ID and one broadcast address, you now have two network IDs and two broadcast addresses, which reduces the total number of available host addresses. Subnetting offers many advantages, but one disadvantage is that you lose possible host addresses each time you divide a network into more subnets.

IPv4 Subnet Calculations Using Formulas

Now you're ready to move on to a more complicated example, performing calculations using formulas and without so much binary involved. Exploring the process in binary helps you understand what's happening "under the hood," so to speak, to clarify the math of what is happening. Realistically, you'll more likely use formulas if you need to calculate subnets on an exam because using formulas typically is a lot faster.

Suppose you want to divide your local network, which has a network ID of 192.168.89.0, into six subnets to correspond to your building's six floors. The following steps guide you through the process:

1. **Decide how many bits to borrow**—How many bits must you borrow from the host portion of the original network ID to get six subnets? Use this formula to determine the number of bits:

$$2^n = Y$$

- n is the number of bits that must be switched from the host address to the network ID.
- Y is the number of subnets that result.

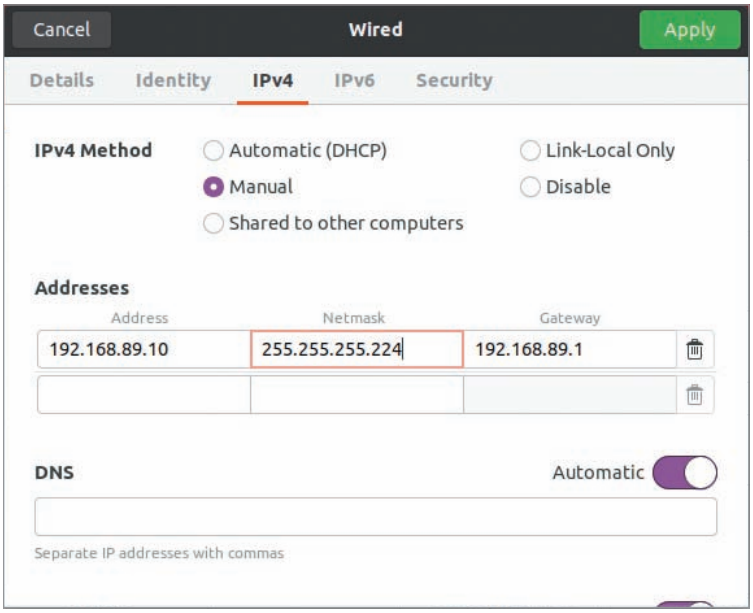
The given scenario requires six separate subnets, meaning that Y, in this case, should be equal to or greater than 6. Experiment with different values for n until you find a value large enough to give you at least the number of subnets you need. For example, $2^2 = 4$; however, 4 is not high enough. Instead consider that $2^3 = 8$; this will give you enough subnets to meet your current needs and allow room for future growth. With 3 being the exponent, replace n in the formula with 3. This tells you that three bits in the host addresses of your class C network must become network ID bits for subnets. This also means that three bits in your subnet mask must change from 0 to 1.

2. **Determine the subnet mask**—The default subnet mask for a class C network is **255.255.255**.0, or **11111111 11111111 11111111** 00000000. In this default subnet mask, the first 24 bits indicate the position of network information.

   Changing three of the default subnet mask's bits from host to network information gives you the subnet mask **11111111 11111111 11111111 111**00000. In this modified subnet mask, the first 27 bits indicate the bits for the network ID. For the original class C network whose network ID is 192.168.89.0, the slash notation for its first subnet would now be 192.168.89.0/27 because 27 bits of the subnet's address are used to provide network information. All other network IDs and IP addresses for the remaining subnets will also require the /27 CIDR block.

   Converting from binary to the more familiar dotted decimal notation, this subnet mask becomes 255.255.255.224 because 11100000 in binary in the final octet is 224 in decimal. When you configure the TCP/IP properties of clients on your network (Figure B-3), you'll specify this new subnet mask.

Figure B-3 IPv4 client configuration for a subnet in Ubuntu Desktop



Source: Canonical Group Limited

Note B-2 | When examining the subnet mask for a network, if any octet is not the usual value of 255 or 0, you know that this network is a subnet and classful addressing is not used. The unusual number (224 in the example) is often called the "interesting octet." Subtract the interesting octet value from 256 and you get what is called the magic number. In this example, the magic number is 256 – 224 = 32. This magic number can be used to calculate the network IDs in all the subnets of the larger network, which you'll practice next.

3. **Calculate the network ID for each subnet**—The first three octets of the network ID for the original class C network (**192.168.89**) are the same for all eight possible subnets. The network IDs differ in the last octet. Use the magic number (as described in Note B-2) to calculate each subnet's network ID as follows:

   - **Subnet 1**: **192.168.89.0**
   - **Subnet 2**: 192.168.89.0 + 32 yields **192.168.89.32**
   - **Subnet 3**: 192.168.89.32 + 32 yields **192.168.89.64**
   - **Subnet 4**: 192.168.89.64 + 32 yields **192.168.89.96**
   - **Subnet 5**: 192.168.89.96 + 32 yields **192.168.89.128**

- **Subnet 6**: 192.168.89.128 + 32 yields **192.168.89.160**
- **Subnet 7**: 192.168.89.160 + 32 yields **192.168.89.192**
- **Subnet 8**: 192.168.89.192 + 32 yields **192.168.89.224**

This method of adding on the same number over and over is called skip-counting. You probably learned this technique in elementary school. For example, skip-counting by twos gives you 0, 2, 4, 6, 8, 10, and so on. Skip-counting by threes gives 0, 3, 6, 9, 12, and so on. In the last octet, you can skip-count by the magic number to get the subnets' final octets. For this scenario, the magic number is 32 and you get the eight subnets' final octets of 0, 32, 64, 96, 128, 160, 192, and 224.

4. **Determine the IP address range for hosts in each subnet**—Recall that you have borrowed three bits from what used to be host information in the IP address. This leaves five bits instead of eight available in the last octet to identify hosts. To calculate the number of possible addresses, keep in mind that each of the five bits has two possible values, a 0 or a 1. Therefore, the number of possible addresses is $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$. But you can't use two of these addresses for hosts because one is used for the subnet's network ID (the one where all five bits are 0 in binary) and one for the broadcast address (the one where all five bits are 1 in binary). This leaves you 30 host addresses in each subnet. As a shortcut to calculating the number of hosts per subnet, you can use the following formula:

$$2^h - 2 = Z$$

- h is the number of bits remaining in the host portion.
- Z is the number of hosts available in each subnet.

In summary, $2^5 - 2$ yields 30 possible hosts per subnet.

In this example, you can have a maximum of $8\,(\text{number of subnets}) \times 30\,(\text{number of hosts per subnet})$, or 240, unique host addresses on the entire, larger network. Recall that each time you subnet a network, you lose two possible host addresses with each new subnet. This overhead is the price you pay for subnetting a network in exchange for the advantages you gain.

When calculating subnets, you'll work with the following information, some of which is initially known and some of which must be calculated:

- Number of subnets
- Number of host addresses per subnet
- Network ID for each subnet
- Broadcast address for each subnet
- Range of possible host addresses within each subnet

In this example, you've already calculated the number of subnets, number of host addresses, and each subnet's network ID. Once you know the network ID of the subnets, calculating the address range of hosts and each subnet's broadcast address is relatively simple. For example, take subnet 5. The network ID is 192.168.89.128. The broadcast address is the last address before the next subnet's network ID. In this case, that's $192.168.89.160 - 1$, which is 192.168.89.159.

For the host addresses in this subnet, start back at the network ID. You won't use the network ID for a host address, so you start with the next value and keep going until you reach the broadcast address for the subnet. Therefore, the available host address range for subnet 5 includes 192.168.89.129 through 192.168.89.158, yielding for this particular subnet a total of 30 available host addresses.

Table B-2 lists the network ID, broadcast address, and the range of usable host addresses for each of the eight subnets in this subnetted class C network. Together, the existing network ID along with the additional bits used for subnet information are sometimes called the extended network prefix.

Table B-2 Subnet information for eight subnets in the sample IPv4 class C network

Subnet number	Network ID (extended network prefix)	Range of host addresses	Broadcast address
1	192.168.89.0 or 11000000 10101000 01011001 00000000	192.168.89.1-30	192.168.89.31 or 11000000 10101000 01011001 00011111
2	192.168.89.32 or 11000000 10101000 01011001 00100000	192.168.89.33-62	192.168.89.63 or 11000000 10101000 01011001 00111111
3	192.168.89.64 or 11000000 10101000 01011001 01000000	192.168.89.65-94	192.168.89.95 or 11000000 10101000 01011001 01011111
4	192.168.89.96 or 11000000 10101000 01011001 01100000	192.168.89.97-126	192.168.89.127 or 11000000 10101000 01011001 01111111
5	192.168.89.128 or 11000000 10101000 01011001 10000000	192.168.89.129-158	192.168.89.159 or 11000000 10101000 01011001 10011111
6	192.168.89.160 or 11000000 10101000 01011001 10100000	192.168.89.161-190	192.168.89.191 or 11000000 10101000 01011001 10111111
7	192.168.89.192 or 11000000 10101000 01011001 11000000	192.168.89.193-222	192.168.89.223 or 11000000 10101000 01011001 11011111
8	192.168.89.224 or 11000000 10101000 01011001 11100000	192.168.89.225-254	192.168.89.255 or 11000000 10101000 01011001 11111111

Note B-3 | You can also calculate the magic number by raising 2 to the power of the number of bits in the host portion of the subnet mask's interesting octet. Use this formula:

$$2^h = \text{magic number}$$

In this example, the host portion has five bits, meaning *h* is 5. Therefore, the magic number is $2^5=32$. You can then use this number to determine the subnets' network IDs.

Several good websites and apps can give you a variety of practice subnet questions so you can become especially comfortable with calculating subnets. Search online for "subnetting practice" or try an app on your phone. A good subnetting practice app will give you options on the types of problems to work with, tell you the correct answers, and explain why those answers are correct.

Calculate IPv4 Subnets and Host IP Address Ranges

Recall that you'll typically find two types of subnet calculation problems on some networking exams:

- Given certain network requirements (such as required number of hosts or required number of subnets), calculate possible subnets and host IP address ranges.
- Given an IP address, determine its subnet's network ID, broadcast address, and first/last host addresses.

Now that you've worked through steps for solving the first type of problem using a class C network ID, you're ready to work through another example of the same type of problem, but this time you'll begin with a class B network ID. Then you'll work through an example of the second type of subnetting problem.

Suppose your organization uses the class B network ID of 172.20.0.0 for its entire network, and you need to create 15 subnets. Complete the following steps, responding to the questions as you go:

1. First decide how many bits to borrow from the host address bits. Recall that you can use the formula $\mathbf{2^n = Y}$. To get at least 15 new subnets (without creating more subnets than necessary), how many bits must be borrowed from the host address portion? How many bits total will be used for identifying a host's subnet?
2. With this information, you can now calculate the subnet mask. The default subnet mask for a class B network is 255.255.0.0, and so the third octet is the one that will change. What is the subnet mask for these subnets, written in dotted decimal notation?
3. The magic number will tell you by what amount to skip-count when you're listing the subnets' network IDs. There are two ways to calculate the magic number: Subtract the interesting octet's value from 256, or use the formula **2h** where *h* is the number of host bits in the interesting octet. What is the magic number you can use to calculate the network IDs?
4. Now you can calculate the network IDs for each subnet. Begin with the original network ID. Then in the third octet, count up by the magic number with each iteration. The last subnet's network ID will be equal to 256 minus the magic number, because you can't use 256 itself in any IP address. What is the CIDR notation for the first subnet's network ID? For the second subnet's network ID? For the last subnet's network ID?
5. If 20 bits are used to identify the network and subnet, this leaves 12 bits to identify each host. Using the formula $\mathbf{2^h - 2 = Z}$ where *h* is the total number of host bits, how many host addresses are possible in each subnet? (You might need a calculator for this step, such as the Windows Calculator app in Scientific mode. Note that it's unusual to see an exam question where you're working with bits in the third octet if you're not allowed to use a calculator during the exam, unless you're taking an advanced networking exam.)
6. The range of available host addresses consists of all the possible IP addresses between the network ID and the broadcast address (which is 1 below the network ID for the next subnet). What is the range of host addresses for the first subnet? For the second subnet? For the last subnet?

Now you're ready to work through the other type of subnetting problem that you'll likely find on some exams: calculating a host's network information.

Calculate an IPv4 Host's Network Information

This time, you'll work backward in your calculations by starting with one host's IP address information. Suppose a server on your network displays the following IPv4 network configuration:

IPv4 address : 192.168.89.130
Subnet mask : 255.255.255.224

Your task is to determine the network ID of the subnet this server is located on, the broadcast address, and the range of available host addresses on this subnet. Complete the following steps, responding to the questions as you go:

1. You don't necessarily need to use binary for these calculations. Find the magic number and go from there. As with your earlier calculations, you need to subtract the interesting octet's value from 256 to get the magic number. What is the magic number?
2. If the interesting octet is located at the end of the subnet mask, you can assume the first three octets of the IP address identify the classful network ID before this network was subnetted. This network ID also serves as the network ID for the first subnet. What is the network ID of the first subnet?

3. You can now use the magic number to calculate the remaining subnets' network IDs. What is the second subnet's network ID? What is the final subnet's network ID?
4. To narrow this down to your server's subnet, either skip-count up from a lower numbered subnet or skip-count down from a higher numbered subnet. Either way will work. You're trying to find a network ID that is as close to the server's IP address as possible without going over. What is the network ID of the server's subnet?
5. You can check the next higher subnet's network ID and subtract 1 to determine the broadcast address of the server's subnet. What is the broadcast address?
6. Finally, any IP address between the subnet's network ID and its broadcast address is the range of available host IP addresses. What is this range?

Subnet Masks and the Logical ANDing Function

To calculate a host's network ID given its IPv4 address and subnet mask, computers follow a logical process of combining bits known as ANDing. In ANDing, a bit with a value of 1 combined, or ANDed, with another bit with a value of 1 results in a 1. A bit with a value of 0 ANDed with any other bit results in a 0. If you think of 1 as "true" and 0 as "false," the logic of ANDing makes sense: ANDing a true statement to a true statement still results in a true statement. But ANDing a true statement to a false statement results in a false statement.

ANDing logic is demonstrated in Table B-3, which provides every possible combination of having a 1 or 0 bit in an IPv4 address or subnet mask.

Table B-3 ANDing

IP address bit	1	1	0	0
Subnet mask bit	1	0	1	0
Resulting bit when combined	**1**	**0**	**0**	**0**

A sample IPv4 host address, its default subnet mask, and its network ID are illustrated in Figure B-4 in both binary and dotted decimal notation. Notice that the address's fourth octet could have been composed of any combination of 1s and 0s, and the network ID's fourth octet would still result in all 0s.

Figure B-4 Example of ANDing a host's network ID

```
        IP address:   11000000  00100010  01011001  01111111    192.  34.  89. 127
AND     Subnet mask:  11111111  11111111  11111111  00000000    255. 255. 255.   0

Equals  Network ID:   11000000  00100010  01011001  00000000    192.  34.  89.   0
```

Figure B-4 shows how ANDing logic is applied to an IPv4 address plus a default subnet mask. It works the same way for networks that are subnetted with classless subnet masks.

Self-Check

1. What is the minimum number of bits that should be borrowed to create 14 subnets?
   a. 4 bits
   b. 2 bits
   c. 16 bits
   d. 8 bits
2. Suppose you're calculating the range of host IP addresses for a subnet (the targeted subnet). If the *next* subnet's network ID is 192.168.42.128, what is the *targeted* subnet's broadcast address?
   a. 192.168.42.96
   b. 192.168.42.0
   c. 192.168.42.127
   d. 192.168.42.255

☐ Check your answers at the end of this appendix.

Solutions to Self-Check Questions

Appendix B: Calculating Subnets

1. What is the minimum number of bits that should be borrowed to create 14 subnets?
   Answer: a. 4 bits
   Explanation: Use the formula $2^n = Y$ to determine how many bits to borrow to create a desired number of subnets, where *n* is the number of bits to borrow and *Y* is the number of subnets that many borrowed bits will create. 2^3 equals 8, which is not enough subnets for the targeted 14 subnets. 2^4 equals 16, which is sufficient for the targeted 14 subnets. Therefore, **4 bits** should be borrowed from the host portion for the network portion.
2. Suppose you're calculating the range of host IP addresses for a subnet (the targeted subnet). If the *next* subnet's network ID is 192.168.42.128, what is the *targeted* subnet's broadcast address?
   Answer: c. 192.168.42.127
   Explanation: The broadcast address is always 1 below the network ID for the next subnet. Therefore, 192.168.42.(128 – 1) yields **192.168.42.127**.